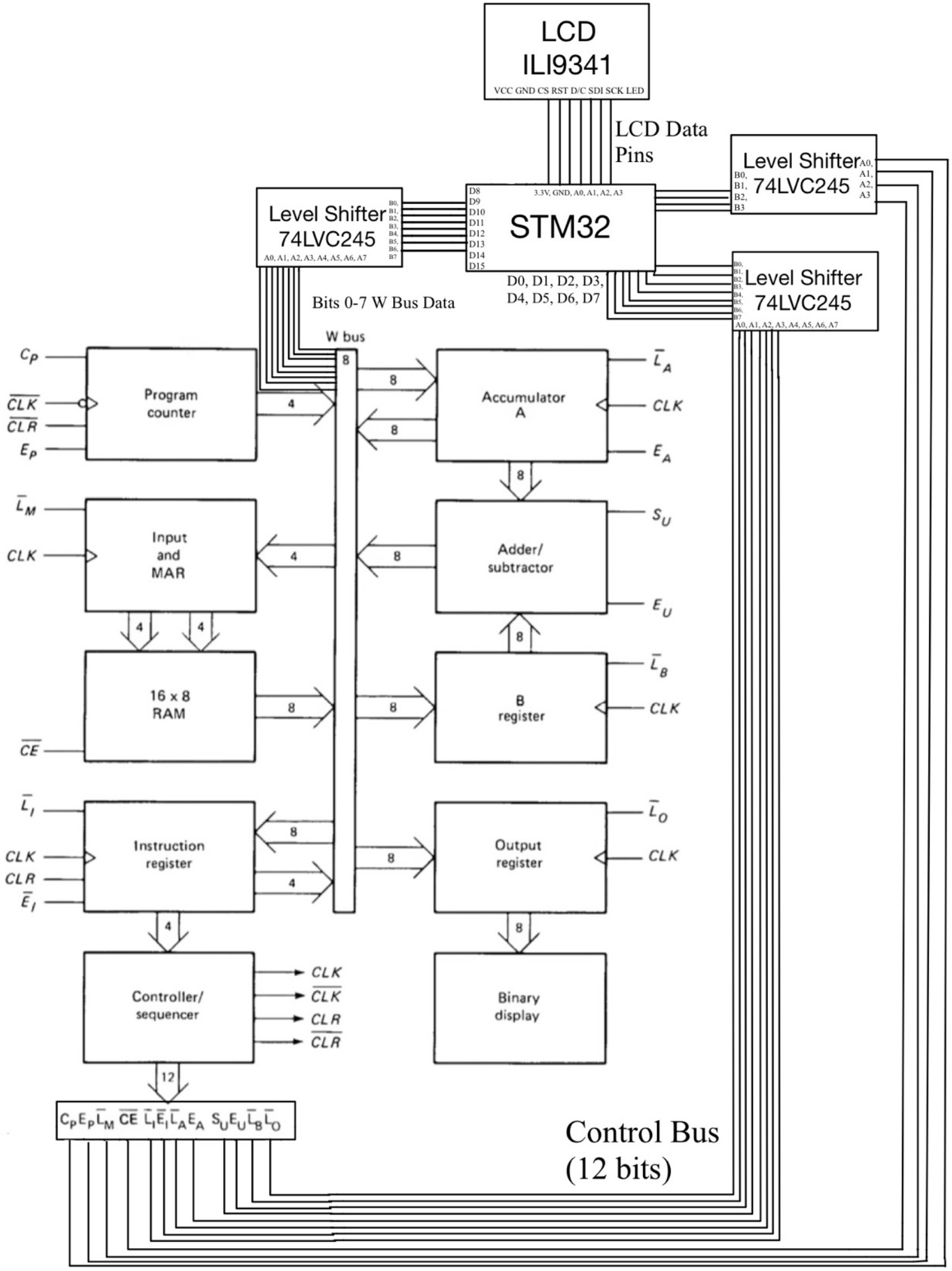
LCD
ILI9341
VCC GND CS RST D/C SDI SCK LED
LCD Data
Pins
Level Shifter
74LVC245
B0, B1, B2, B3
A0, A1, A2, A3
Level Shifter
74LVC245
A0, A1, A2, A3, A4, A5, A6, A7
B0, B1, B2, B3, B4, B5, B6, B7
D8 D9 D10 D11 D12 D13 D14 D15
3.3V, GND, A0, A1, A2, A3
STM32
D0, D1, D2, D3,
D4, D5, D6, D7
Level Shifter
74LVC245
B0, B1, B2, B3, B4, B5, B6, B7
A0, A1, A2, A3, A4, A5, A6, A7
Bits 0-7 W Bus Data
W bus
Program counter
Accumulator A
Input and MAR
Adder/ subtractor
16 x 8 RAM
B register
Instruction register
Output register
Controller/ sequencer
Binary display
Control Bus
(12 bits)